

“I used ChatGPT to review my draft explanation”

IRWA Project Part-2: Indexing and Evaluation

Authors: Carla Núñez u214970 / Julia Pérez u214026 / Jan Prats u213927

GitHub URL: https://github.com/Januto30/SearchEngine-IRWA_2025

TAG: IRWA-2025-part-2

PART 1: Indexing

1.1. Inverted Index and TF-DF Construction

To enable efficient document retrieval and relevance ranking, we have made an inverted index from the product dataset. Each product entry contained preprocessed text attributes, including the title, description, and details, which were concatenated into a single textual representation for indexing.

From this combined text, the system tokenized terms and stored them in an inverted index structure mapping each term to a list of documents (identified by doc_id) and the positions where the term appears. In parallel, the system computed:

- **Term Frequency (TF):** normalized by document length to ensure fair weighting across variable-length texts,
- **Document Frequency (DF):** counting the number of documents in which a term appears, and
- **Inverse Document Frequency (IDF):** calculated as $\text{idf}(t) = \log((1 + N) / (1 + \text{df}_t))$, where N is the total number of documents and df_t is the document frequency of term t.

The resulting data structures (index, tf, df_counts, idf, and doc_titles) collectively form the TF-IDF model, allowing retrieval of documents ranked by relevance to a given query.

The retrieval model supports **conjunctive (AND) queries**, meaning that only documents containing all query terms are considered matches. For each query:

1. The system identifies all documents containing each query term using the inverted index.
2. It computes the intersection of these document sets to enforce the AND condition.
3. Each resulting document is then assigned a TF-IDF score, calculated by summing the product of its TF and IDF values for all query terms.
4. Documents are finally ranked in descending order of TF-IDF score to present the most relevant results first.

A mapping from internal document identifiers to product IDs (pid_map) ensures that ranked results correspond directly to products in the original dataset.

To validate the system, several representative queries were executed on the indexed dataset:

Query	Top Results	Observation
hardware components	No results found	The dataset does not contain technical hardware products.
round neck blue shirt	Multiple T-shirts with high scores (1.38–1.44)	High relevance; strong keyword overlap in titles and descriptions.
casual shirt men	Several men’s casual shirts (scores $\approx$ 0.84–0.86)	Consistent ranking, showing proper TF-IDF weighting.
women slim fit	Women’s slim-fit shirts (scores $\approx$ 1.28–1.42)	Effective retrieval of gender and fit-specific items.
polo ice cream	No results found	Query terms do not co-occur in any product entry.

Overall, the system demonstrates effective retrieval and ranking behavior for well-represented terms in the dataset. The absence of results for certain queries reflects dataset domain limitations rather than retrieval inefficiency.

PART 2: Evaluation

The second part of the project focused on assessing the performance of our retrieval system based on the inverted index and TF-IDF ranking implemented in Part 1.

The goal was to evaluate how effectively the system retrieves relevant documents for different user queries, using standard Information Retrieval (IR) metrics.

To do this, we first implemented a set of evaluation functions and then applied them to both predefined validation queries (with external relevance labels) and our own custom test queries from Part 1.

1. Implementation of Evaluation Metrics

We implemented all the required evaluation metrics as separate Python functions:

Precision@K (P@K): Measures the proportion of retrieved documents among the top K that are relevant.

- **Recall@K (R@K):** Computes the fraction of relevant documents retrieved among all relevant documents in the collection.
- **Average Precision@K (AP@K):** Averages the precision values across ranks where relevant documents appear.
- **F1-Score@K:** Harmonic mean between Precision and Recall, balancing both metrics.
- **Mean Average Precision (MAP):** Average of AP@K over all queries, representing global retrieval quality.
- **Mean Reciprocal Rank (MRR):** Evaluates how early the first relevant result appears.

- **Normalized Discounted Cumulative Gain (NDCG)**: Takes into account the ranking position of relevant documents, giving higher importance to top-ranked items.

To streamline the process, we designed an additional helper function called `evaluate_query()`. This function integrates all the above metrics and returns them in a single dictionary, rounded to three decimal places as requested.

2. Evaluation with Predefined Validation Queries

The first evaluation used the two predefined validation queries supplied in the project dataset:

1. *"women full sleeve sweatshirt cotton"*
2. *"men slim jeans blue"*

Each query was compared against the ground-truth labels in `validation_labels.csv`.

All metrics (Precision, Recall, MAP, NDCG, etc.) resulted in 0.000, meaning no relevant products were retrieved.

We believe the main cause lies in our conjunctive query model, which requires that a product contain *all* query terms simultaneously. Since these validation queries are relatively long and specific, it is unlikely that any product title or description includes every single term.

Consequently, the strict Boolean AND condition drastically reduces recall. If we had used a disjunctive (OR) model retrieving documents that contain *any* of the query terms or introduced term weighting, emphasizing key terms with higher discriminative power, the system might have retrieved partial but still relevant matches. We strongly think that these strategies would work better on long queries instead of the conjunctive one.

3. Establishing Ground Truth for Custom Queries & Evaluation

Next, we manually defined binary relevance judgments for our five custom queries from Part 1:

- *"hardware components"*
- *"round neck blue shirt"*
- *"casual shirt men"*
- *"women slim fit"*
- *"polo ice cream"*

Each retrieved document was labeled as 1 (**relevant**) or 0 (**non-relevant**) based on a manual inspection of its title and context. This manual evaluation acts as the expert judgment required by the project, enabling a fairer qualitative assessment of our own queries.

The system generated the top five ranked documents for each query. Two of the queries, *hardware components* and *polo ice cream*, produced **no retrieved documents**, suggesting limited dataset coverage.

For the remaining three queries, **all retrieved documents were judged relevant**, reflecting a strong alignment between query intent and document content:

- **Query 2: “round neck blue shirt”** → 5/5 relevant
- **Query 3: “casual shirt men”** → 5/5 relevant
- **Query 4: “women slim fit”** → 5/5 relevant

After establishing the ground truth, several standard Information Retrieval metrics were computed for each query:

Query	P@K	R@K	AP@K	F1@K	RR	NDCG
Hardware components	—	—	—	—	—	—
Round neck blue shirt	1.0	1.0	1.0	1.0	1.0	1.0
Casual shirt men	1.0	1.0	1.0	1.0	1.0	1.0
Women slim fit	1.0	1.0	1.0	1.0	1.0	1.0
Polo ice cream	—	—	—	—	—	—

The aggregated metrics across all five queries were:

- **MAP = 1.000**
- **MRR = 1.000**

These perfect scores indicate that the system successfully retrieved and ranked only relevant documents for all queries that had matching results in the dataset.

The results demonstrate excellent retrieval accuracy and ranking quality for the three queries where relevant documents existed. This suggests that the search and ranking functions are effectively capturing term relevance, product descriptions, and user intent. However, the two queries with no retrieved results (*hardware components* and *polo ice cream*) highlight potential limitations in:

- The **coverage** or domain representation of the dataset.
- The **query-document vocabulary alignment**, which could be improved with synonym expansion or semantic retrieval techniques.

Overall, the evaluation confirms that the current retrieval model performs optimally for represented product categories such as apparel, achieving perfect precision, recall, and ranking metrics.